# Alejandro Pelaez-Duque

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## EDUCATION

**Brigham Young University**, Ira A. Fulton College of Engineering, Provo, Utah**:** Apr 2027
Bachelor of Science, *Computer Engineering*, Research Focus in Autonomous Aerial & Ground Systems
Minors in Mathematics and Computer Science **GPA: 3.87**

- Relevant coursework: **Graduate level** Autonomous Aircraft, High-Performance Computing. **Senior level** Control Systems, Robotics & Autonomy, Applied Machine Learning.

## PUBLICATIONS & PRESENTATIONS

### Publications

1. **Alejandro Pelaez**, Randal W. Beard, "COAST: Safe Flight Corridor Planner for Fast and Robust Snap Minimized Trajectories," *2027 American Control Conference (ACC).* [Submitted October 2026]
2. **Alejandro Pelaez**, Dallon Gunn, Brady Moon, "What Transfers to Road Topology? A Controlled Study of Foundation-Model Representations Across Architectures and Geographic Domains," Manuscript in preparation
3. Curtis C. Johnson, Carlo Alessi, Egidio Falotico, **Alejandro Pelaez**, Cameron Collyer, Randon Sandall, Marc D. Killpack, "Zero-shot Whole-Body Manipulation with a Large-Scale Soft Robotic Torso via Guided Reinforcement Learning," Manuscript in preparation
4. Muhammad Taha, Caleb Fears, **Alejandro Pelaez**, Gregory Nordin, Adam Woolley, Troy Munro, "3D Printed, Peltier Sensor for Measurements Within Small Scale Thermal Systems," *International Thermal Conductivity Conference & International Thermal Expansion Symposium (ITCC/ITES 2026)*. [Accepted and presented by Muhammad Taha; not yet published]
5. Muhammad Taha, Mitchell Martin, Caleb Fears, Nathan Smith, **Alejandro Pelaez**, Adam Woolley, Gregory Nordin, Troy Munro, "Development of Miniaturized Thermoelectric Element for Temperature Control in Microfluidic Systems," Manuscript in preparation

### Research Presentations

1. **Alejandro Pelaez**, Randal W. Beard, "COAST: Real-Time Autonomous Aerial Planner with Safety Guarantees for Fast and Robust Trajectories" Poster Presentation, 2026 SHPE National Convention, Indianapolis, October 2026. **(2nd Place Podium, Undergraduate Poster Competition)**
2. **Alejandro Pelaez**, "Safe Flight Corridor Planner for Fast and Robust Snap Minimized Trajectories," Oral Presentation, *BYU IMMERSE*, Brigham Young University, Provo, UT, August 2026.
3. **Alejandro Pelaez**, "Self-Clearing Channels," Oral Presentation, *BYU IMMERSE*, Brigham Young University, Provo, UT, May 2024.

## RESEARCH EXPERIENCE

**Autonomous Aerial Systems Research Assistant** *Feb 2026-Present*
*BYU MAGICC (Multiple Agent Intelligent Coordination and Control) Lab, Dr. Randy Beard (RPI Alum)* - Provo, UT

- Engineered a C++ online trajectory planner using KD-Tree spatial projections and B-Spline kinodynamic bounding, achieving **sub-10ms computation times** for real-time autonomous UAV navigation.
- Integrated simulated LiDAR point clouds with OSQP-based Quadratic Program within a RViz environment, ensuring dynamically feasible, obstacle-avoiding flight paths through continuous-space Safe Flight Corridors.
- **Authored manuscript** on snap-minimized trajectory optimization submitted to the American Control Conference (ACC), demonstrating the system's speed and advantages over state-of-the-art planners.

**Computer Vision & Neural Networks Research Assistant** *Aug 2025-Present*
*BYU FROST (Field Robotic Systems) Lab, Dr. Brady Moon (CMU Alum)* - Provo, UT

- Spearheaded a systematic empirical comparison of vision foundation models (SAM 2.1, DINOv3, and C-RADIOv3) for satellite road network extraction, culminating in a first-author journal manuscript (in preparation).
- Evaluated architectural augmentations and cross-domain generalization across 3,300+ global satellite images, demonstrating that SAM 2.1 improves out-of-domain topological routing (APLS) by over 4 points compared to prior state-of-the-art baselines.
- Engineered a controlled Python evaluation pipeline on an **NVIDIA RTX PRO 6000** GPU to isolate encoder features, proving that foundation-model pretraining paradigms impact graph connectivity significantly more than downstream decoder complexity.

**Robotics and Control Research Assistant** *Sep 2026-Present*
*BYU RAD (Robotics and Dynamics) Lab, Dr. Marc Killpack (Georgia Tech Alum)* - Provo, UT

- Developed evaluation methodology for RL-based controller against PID and MPC methods.
- Tested and tuned MPC controller and cost function in 1, 3, and 6 DOF scenarios.

**Microfluidics and Materials Science Research Assistant** Feb 2024-Jun 2025

*BYU Electrical Engineering, Dr. Greg Nordin (USC Alum) & Dr. Troy Munro (KU Leuven/USU Alum)* - Provo, UT

- Collaborated with Mechanical Engineering department towards progress of conference and journal papers by providing microfluidic 3D prints within 3 days of print request and within 1 week of design request.
- Mentored **2 undergraduate students** towards effective implementation of plans for research project and best practices by holding weekly accountability meetings and goal planning sessions.
- Developed electronic automation platform to streamline device preparation and reduce human error. Removed human preparation and sped up process from **2 days to 4-6 hours**.

## PROFESSIONAL EXPERIENCE

**Software Engineer Intern** *Jun 2025-Aug 2025*

*Bank of New York (BNY) Mellon* - Pittsburgh, PA

- Collaborated with business team to develop proprietary OpenAI based agent (BNY's ELIZA) for team needs, saving over **$150k** in resources through redirecting user traffic to AI agent instead of business team.
- Leveraged MS Power BI API and SQL databases to create financial report for high value user access, decreasing process time by **52%** through streamlining UX.

**Electromagnetism and Optics Laboratory Assistant** *Jan 2024-May 2024*

*BYU Department of Physics and Astronomy* - Provo, UT

- Mentored 56 students in proper understanding of electromagnetic phenomena and elemental circuitry.
- Demonstrated lab procedures through detailed explanation and examples leading to 100% completion rate.

**Licensed Commercial Electrician** *May 2022-Feb 2024*

*W Systems Electrical Contracting* - Draper, UT

- Motivated coworkers to increase performance output by producing 80% of assigned units, incident-free, following OSHA guidelines despite being 1 of 4 workers within group.
- Entrusted with testing and troubleshooting to ensure quality electric layout and proper functionality of over 15 apartment units.

## TECHNICAL PROJECTS

**Autonomous Robotic Fleet - Independent**

- Developing fleet of cars from scratch. Drafted circuitry sketch and sensor connections (distance sensors, IMU, servos).
- Prototyped circuit in breadboard and tested functionality with voltmeter and sensor mock code.
- Designed custom two layered PCB in KiCad. Learned Fusion 360 and designed car enclosure to exact fitting of components.
- Iterated through 3D prints until final product was up to measurements. Currently learning to export URDF file into python with goal of simulating environments for RL training (PyBullet, Pytorch).

**RC Wi-Fi Based Communication Vehicle - Independent**

- Used ESP32's as TX/RX systems. One ESP acted as the user input, while the other actuated sensor signals within RC car.
- Utilized pointers, memory allocation, and FSMs to reduce latency time.
- Learned Wi-Fi networking protocols (MAC address connections, and packet sending and receiving) to connect ESPs together. Reduced latency time to <=100ms.

**Self-Leveling Drone - Independent**

- Built, soldered, and tested a drone with use of a transmitter/receiver controller and a gyroscope/accelerometer to detect changes in orientation and tilt (C++/Arduino).
- Learned drone architecture (microcontroller -> ESC -> brushless motors) and basic control systems through integrating a gyroscope to control motor speed and drone tilt using PID control.

## LEADERSHIP EXPERIENCE

**Society of Hispanic Professional Engineers Chapter Leadership: Vice-President, President, Secretary** Apr 2023-Present

- Directed fundraising efforts of **$60k+** for helping members attend yearly national convention.
- Coordinated travel itineraries with university's travel office for 50+ students
- Grew club membership by **35%** through pitching club benefits at seminars totaling over 400 people.

**Young Men's Advisor** Jan 2023-Jan 2024

- Aided in development of over 20 low-income immigrant and multicultural young men in social and academic settings. Led by example in showing youth how to take advantage of academic opportunities.
- Organized youth camps tailored to needs of youth. Funded these experiences through city-wide fundraisers.

## HONORS AND AWARDS

- Placed **second nationally** in SHPE's National Convention undergraduate research competition – Poster.
- Placed **third nationally** in IBM/Qualcomm sponsored jeopardy-style national STEM competition with over 200 competitors.
- **HSF Scholar**. Accepted to the Hispanic Scholarship Fund, a prestigious program with 14% acceptance rate.
- BYU Scholarship recipient (2022-present) for High Academic performance.

## SKILLS

**Languages:** Python, C/C++, MATLAB
**ML/Robotics:** PyTorch, OpenCV, ROS, PyBullet, IsaacSim
**Embedded:** Arduino, ESP32, Raspberry Pi
**Design:** KiCad, Fusion 360, OpenSCAD
**Systems:** Docker, Apptainer, Git, Linux